

Research Prospect Finder

Product Requirements Document (PRD) — CLI-first, AI-minimal, web-ready

Purpose: Personal tool for discovering established local businesses that may become research/skripsi objects, analyzing research opportunities, generating topic candidates, and managing respectful outreach.

Core principle: deterministic automation first; AI only when reasoning is needed; human validation remains mandatory.

1. Product Overview

- Interactive CLI application for finding established local businesses within a chosen area.
- Collects public business signals, filters small/irrelevant businesses, ranks candidates, optionally analyzes selected candidates with AI, generates research opportunities and topic ideas, and manages email outreach.
- Designed so the same application/domain core can later be exposed through FastAPI and a web frontend.

2. Problem Statement

- Finding a suitable skripsi object is slow because business discovery, contact research, operational-problem discovery, note taking, and outreach are performed manually.
- The product turns this into: Discover → Filter → Rank → Analyze → Select → Contact → Validate.

3. Goals

- Find established local businesses rather than very small informal sellers.
- Produce a shortlist of research candidates.
- Minimize AI calls and token consumption.
- Cache AI results and avoid repeated analysis.
- Support manual review before outreach.
- Keep the core independent from the CLI so a future web app does not require rewriting business logic.

4. Non-Goals

- Not a mass-marketing platform.
- Not a fully autonomous sales agent.
- Not a replacement for interviews or field validation.
- Not a tool for accessing private/non-public information or bypassing access controls.

5. Target Businesses

- Examples: established local workshops, clinics, restaurants, cafes, multi-outlet laundry businesses, printers, distributors, furniture/electronics stores, suppliers, gyms, pet shops, education businesses, travel agencies, wedding organizers, and garment/convection businesses.
- Exclude or deprioritize tiny home businesses, individual sellers, large national corporations, and large franchises.

6. User Journey

- Launch interactive CLI → choose location → choose radius → choose categories → configure business-size filters → discover → review/rank → save candidates → optionally audit online presence → analyze top candidates with AI → inspect opportunities → generate topic candidates → draft/review email → send with approval → track outreach.

7. Interactive CLI

- Main menu: Discover Businesses; Saved Businesses; Analyze Prospects; Research Opportunities; Research Topics; Outreach; Dashboard; Settings; Exit.
- Use interactive selection, multi-select filters, progress indicators, tables, keyboard navigation where practical, and clear error/retry states.
- Also provide non-interactive commands for automation and scripting.

8. Discovery & Filtering

- Inputs: current location/address/coordinates, radius, categories, minimum rating, minimum review count, online-presence preference, local-business preference, franchise preference.
- Pipeline: provider search → deduplication → basic data collection → deterministic filtering → business-size score → online-presence checks → candidate ranking.
- Review count is only a signal, not proof of company size.

9. Business Data

- Store business name, category, address, latitude/longitude, phone, website, rating, review count, hours, branches when available, source, timestamps, and status.
- Store public online-presence signals such as website and public social profiles.

10. Scoring

- Initial deterministic score can combine business-size signal, online presence, customer/review signal, operational complexity, research accessibility, and contact availability.
- Example weights: size 25%, online presence 15%, customer signal 15%, operational complexity 20%, research accessibility 15%, contact availability 10%. Make weights configurable.

11. Website / Online Presence Analysis

- Use ordinary HTTP extraction and rules before AI.
- Collect title, description, services, contact details, booking/order features, customer portal signals, forms, e-commerce signals, technology indicators, navigation/usability signals, and social links where publicly available.

12. AI Strategy

- AI is optional and should only run after deterministic filtering/ranking.
- Recommended flow: hundreds of businesses → rules → shortlist → top 5–20 → one AI request per selected business.
- One AI request should return potential problems, information-system opportunities, research relevance, topic candidates, and validation questions.
- Cache results using business/data hashes; re-run only for new/changed data or explicit --reanalyze.
- Provide a daily AI-analysis limit and a disabled-AI mode.

13. AI Output

- AI should phrase findings as hypotheses based on public evidence, not facts about the business.
- Output: potential operational problems, potential information-system opportunities, confidence/reasoning, 3–5 topic candidates, and questions to validate with the owner.

14. Research Topics

- Each topic should include title, problem, proposed system, target users, scope, and validation questions.
- Topics are suggestions; the user validates the real problem through interview/observation.

15. Outreach

- Use public business contact information intended for business contact.
- Default to deterministic email templates; AI personalization is optional.
- Require human review and approval before sending in the MVP.
- Track statuses: NOT_CONTACTED, DRAFT, READY, SENT, DELIVERED, REPLIED, INTERESTED, DECLINED, NO_RESPONSE, DO_NOT_CONTACT.
- Respect requests not to be contacted.

16. Database

- Use SQLite for MVP and local-first operation.

- Core entities: businesses, business_sources, online_presence, business_scores, ai_analyses, research_opportunities, research_topics, contacts, email_drafts, outreach_events, settings.

17. Export & Dashboard

- Export CSV, JSON, and Markdown.
- Dashboard metrics: discovered, qualified, saved, analyzed, opportunities, topics, sent, replies, interested, no response.

18. Providers & Architecture

- Use provider abstractions for business search, website access, AI, and email.
- AI provider interface should support OpenAI-compatible endpoints, 9Router-compatible configuration, and a future local model.
- Keep CLI as an adapter over application/domain services.

19. Recommended Stack

- Python; Typer; Rich; Pydantic; SQLAlchemy; SQLite; httpx; BeautifulSoup.
- Future web: FastAPI + PostgreSQL + optional Redis + existing core services.

20. Project Structure

- `research-finder/src/research_finder/{cli,application,domain,providers,database,services,config};`
`data/{research.db,cache,exports}; tests/.`

21. MVP Phases

- Phase 1 Foundation: project, CLI, interactive menu, SQLite, configuration.
- Phase 2 Discovery: location, radius, category, provider search, deduplication, filters, persistence.
- Phase 3 Ranking: deterministic score, ranking, candidate selection.
- Phase 4 Website Analysis: website discovery and rule-based online presence audit.
- Phase 5 AI: provider abstraction, 9Router/OpenAI-compatible support, caching, limits, analysis, opportunities, topics.
- Phase 6 Outreach: contact discovery, templates, draft/review/send/tracking.
- Phase 7 Polish: dashboard, export, logging, errors, backup.

22. Success Criteria

- User can discover and filter businesses interactively, save candidates, analyze only selected prospects, obtain research opportunities and topic candidates, draft/review/send outreach, and track responses without repeated AI calls.
- All core functionality remains usable when AI is disabled.

23. Safety / Data Principles

- Use public business information and normal provider access.
- Do not collect private data, bypass authentication, or access non-public areas.
- Use rate limiting, retries, backoff, caching, and provider quotas.
- Outreach must be reviewable and stoppable.

Appendix — Example CLI Flow

```
$ research-finder
? What do you want to do?
> Discover businesses
Saved businesses
Analyze prospects
Research opportunities
Research topics
Outreach
Dashboard
Settings
Exit
```

Discovery should end with a ranked shortlist and allow the user to select which businesses consume AI budget.